Design Issues in MIPS Instruction Pipeline

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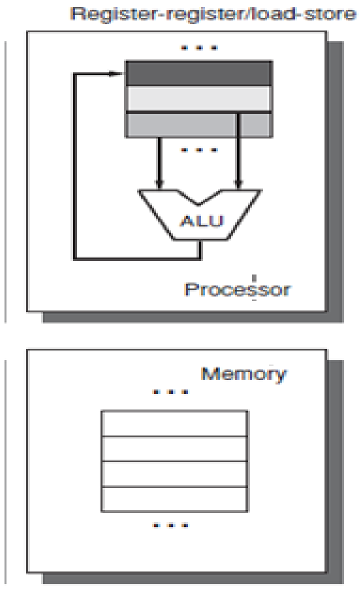
Introduction to MIPS

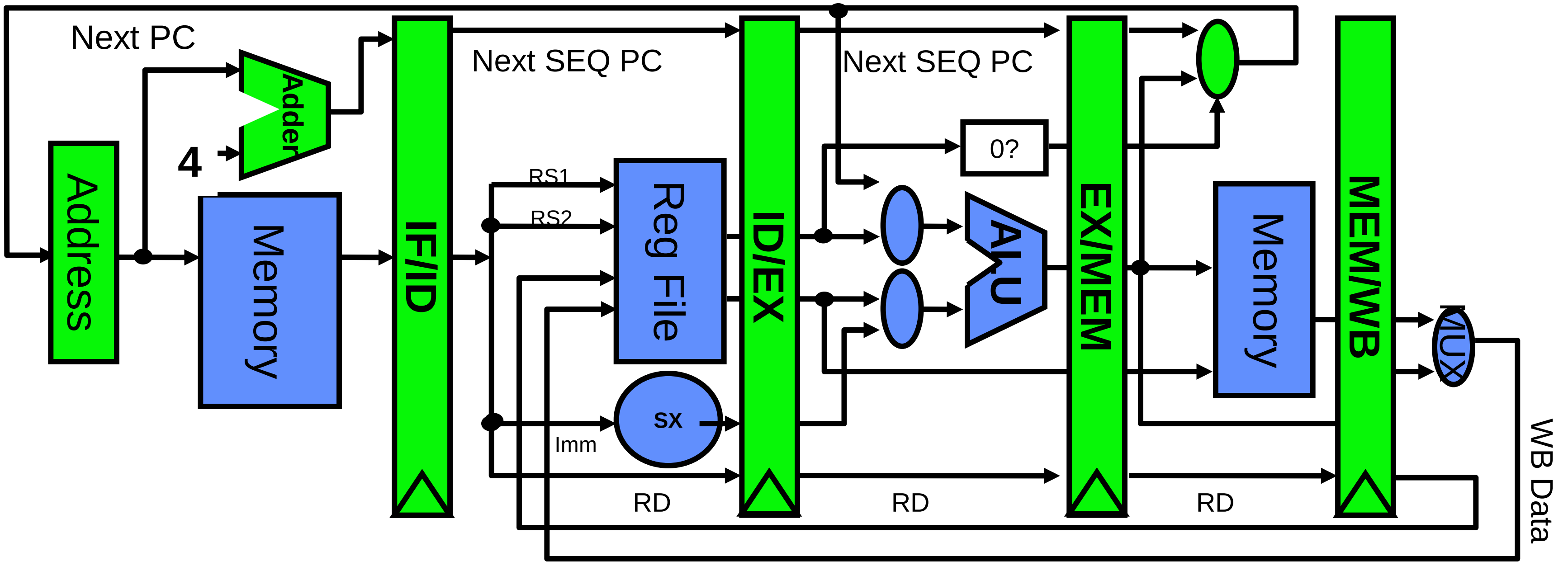
- ❖Microprocessor without Interlocked Pipelined Stages
- ❖32 registers (32 bit each)
- ❖Uniform length instructions
- ❖RISC- Load store architecture

	6 bits	5 bits	5 bits	5 bits	5 bits	6 bits
R:	op	rs	rt	rd	shamt	funct
I:	op	rs	rt	address / immediate		
J:	op	target address				



Pipelined RISC Data path

Instruction Fetch	Instr. Decode Reg. Fetch	Execute Addr. Calc	Memory Access	Write Back



5 Steps of RISC Data path

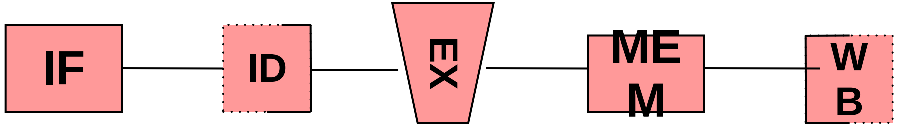
- Each instruction can take at most 5 clock cycles
- **Instruction fetch cycle (IF)**
  - Based on PC value, fetch the instruction from memory
  - Update PC
- **Instruction decode/register fetch cycle (ID)**
  - Decode the instruction + register read operation
  - Fixed field decoding
  - Equality check of registers
  - Computation of branch target address if any

5 Steps of MIPS Data path

- **Execution/Effective address cycle (EX)**
  - Memory reference: Calculate the effective address
  - Register-register ALU instruction
  - Register-immediate ALU instruction
- **Memory access cycle (MEM)**
  - Load instruction: Read from memory using effective address
  - Store instruction: Write the data in theregister to memory using effective address

5 Steps of MIPS Data path

- **Write-back cycle (WB)**
  - Register-register ALU instruction or load instruction
  - Write the result into the register file
- **Cycles required to implement different instructions**
  - Branch instructions – 4 cycles
  - Store instructions – 4 cycles
  - All other instructions – 5 cycles



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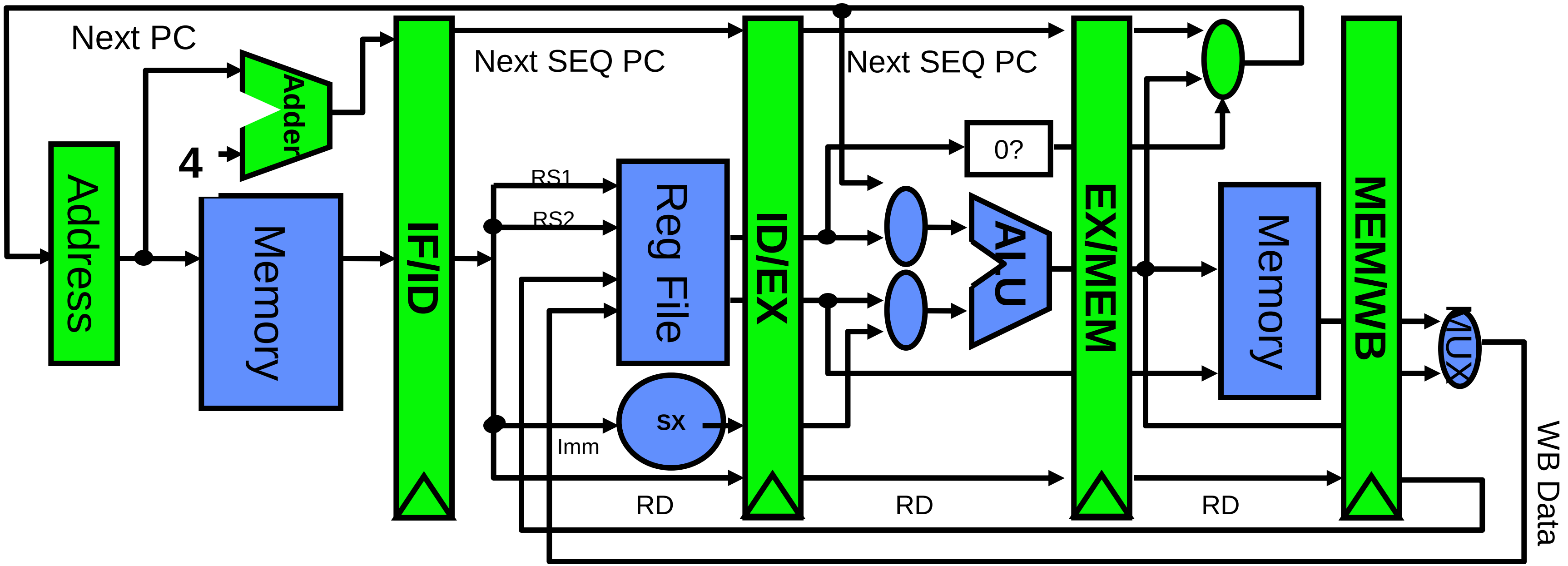
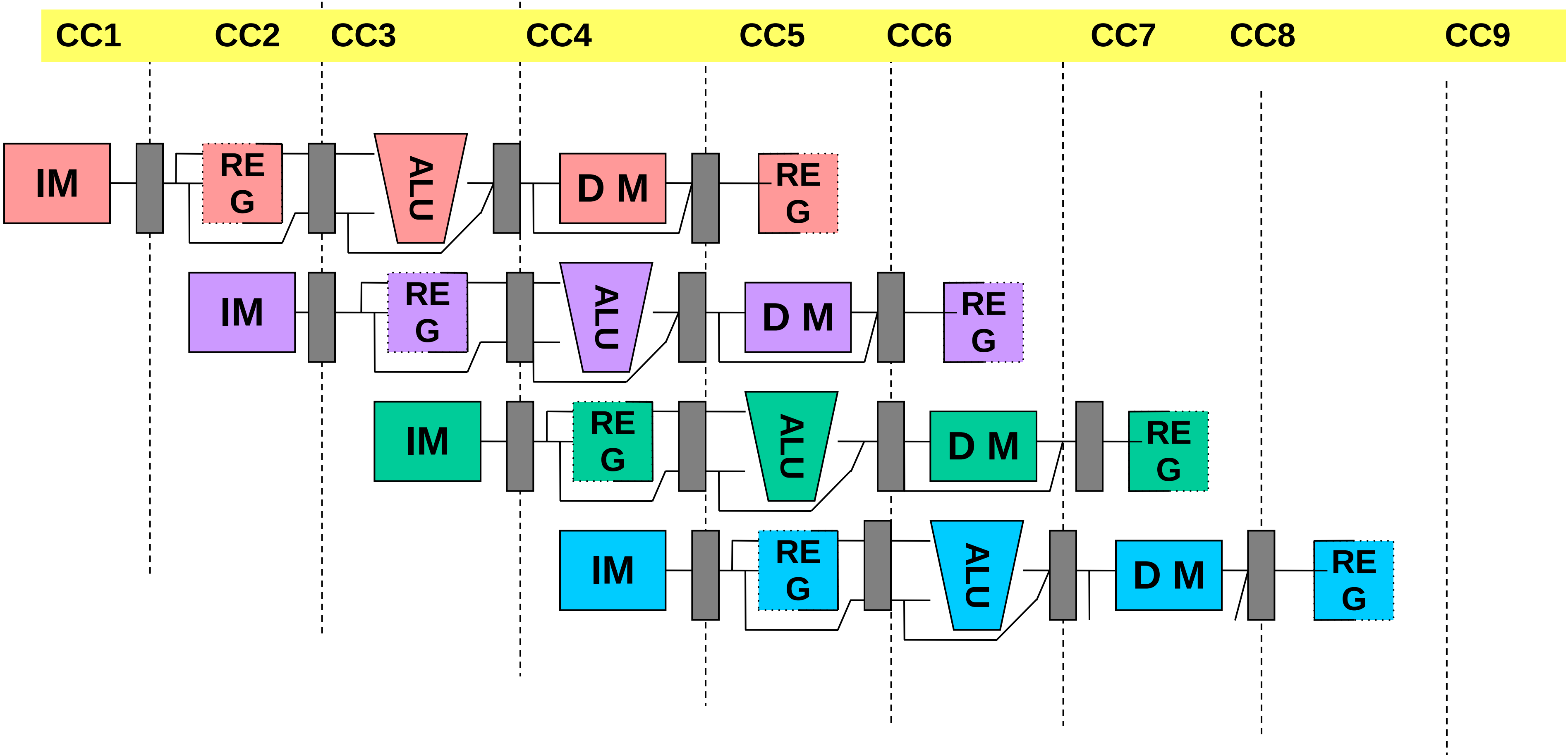


Visualizing Pipelining

Clock number								
Instruction number	1	2	3	4	5	6	7	8
i	IF	ID	EX	MEM	WB			
i+1		IF	ID	EX	MEM	WB		
i+2			IF	ID	EX	MEM	WB	
i+3				IF	ID	EX	MEM	WB
i+4					IF	ID	EX	MEM

Visualizing Pipelining



Pipelining Issues

- **Ideal Case: Uniform sub-computations**
  - The computation to be performed can be evenly partitioned into uniform-latency sub-computations
- **Reality: Internal fragmentation**
  - Not all pipeline stages may have the uniform latencies
- **Impact of ISA**
  - Memory access is a critical sub-computation
  - Memory addressing modes should be minimized
  - Fast cache memories should be employed

Pipelining Issues

- ❖ **Ideal Case : Identical computations**
  - ❖ The same computation is to be performed repeatedly on a large number of input data sets
- ❖ **Reality: External fragmentation**
  - ❖ Some pipeline stages may not be used
- ❖ **Impact of ISA**
  - ❖ Reduce the complexity and diversity of the instruction types
  - ❖ RISC architectures use uniform stage simple instructions

Pipelining Issues

- **Ideal Case : Independent computations**
  - All the instructions are mutually independent
- **Reality: Pipeline stalls – cannot proceed.**
  - A later computation may require the result of an earlier computation
- **Impact of ISA**
  - Reduce Memory addressing modes - dependency detection
  - Use register addressing mode - easy dependencies check

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